

Numerical study on satellite droplet formation in laminar jet breakup for a liquid droplet radiator



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ABSTRACT

Liquid Droplet Radiator is promising for the waste heat removal of space nuclear power systems, and droplet properties have an impact on its performance. Breakup behaviors of laminar liquid jets with continuous droplets formation were numerically studied. Results verified that satellite droplets formation would destroy uniform droplet streams. The effects of We , Oh , Ca , and k on laminar jet breakup behaviors were investigated. Increasing Oh shifted jet breakup pattern from downstream pinching to upstream pinching, and it reversed as We increased. The jet breakup was more delayed with Oh increased at low We and jet oscillation amplitude became larger with We increased. At low We , breakup length and breakup time decreased firstly and then increased with k increased, but increased with Ca increased. However, the influence of Ca became slight at high We . Moreover, quantitative criteria were proposed for accurately identifying three satellite droplet formation regimes.

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1. Introduction

With the rapid development of deep space exploration and space station construction, the more consistently reliable energy and power supply for large space structures are urgently needed. The space nuclear power system can provide both electricity and propulsion for the spacecraft, which will play an unparalleled role in future aerospace engineering due to its longer lifetime and lower power-to-weight ratio compared to the traditional energy sources (Jian et al., 2020; Qin et al., 2020). It is well known that convective heat transfer is almost impossible to achieve in space and hence the heat dissipation of space nuclear power systems can be accomplished only through radiative heat transfer. Compared to traditional space heat removal system, the liquid droplet radiator (LDR) is a promising heat removal equipment possessing a series of outstanding advantages, such as low weight, small volume, high heat dissipation efficiency, strong safety, and long lifetime (Massardo et al., 1997; Mattick and Hertzberg, 1982; Tagliafico and Fossa, 1997; Ye and Ma, 2011; Koroteev et al., 2016; Huang et al., 2021; Yang et al., 2021; Qin et al., 2019, 2020, 2021). The schematic diagram of an LDR configuration is shown in Fig. 1, which mainly contains a droplet generator, a droplet layer, and a droplet collector. The droplet layer is used as the effective heat transfer region of LDR, which is supposed to be filled with a cloud

of tiny liquid droplets with uniform size and velocity. Consequently, the working performance of an LDR is determined by the heat dissipation characteristics of the droplet layer (Ye and Ma, 2011; Koroteev et al., 2016; Qin et al., 2019; Yang et al., 2021).

The heat dissipation process of an LDR involving radiation heat transfer and evaporation of working fluid was mainly investigated by numerical calculation in the past research. Ye and Ma (2011) developed the combined radiation-evaporation model of a rectangular droplet layer in space and found that the optical thickness of the droplet layer related to the droplet distribution plays an important role in both radiation heat transfer and evaporation loss. Huang et al. (2021) systematically investigated the influence of droplet radius, droplet velocity, initial temperature, and working medium on the radiation heat transfer characteristics of LDR. There is no doubt that the droplet distribution characteristics and droplet size have a considerable impact on the heat and mass transfer of the droplet layer. The previous studies demonstrated that the formation characteristics of droplets have important significance for the working performance of an LDR. In consideration of energy and mass transfer existing, it is obtained that the temperature distribution of liquid jet and droplet stream was almost constant within a short distance and the breakup process of the liquid jet was mainly governed by the momentum exchange rather than the energy exchange (Li et al., 2007; Yong et al., 2014). Chong et al. (2021) summarized that the evaporation loss of DC 705 silicone oil and molten tin for an LDR with a 30-year lifetime, and was very small. It indicates that the influence of heat and mass

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Nomenclature

B_0	Bond number
Ca	Capillary number
D	diameter
D_d	droplet diameter (μm)
D_j	jet diameter (μm)
D_0	nozzle diameter (μm)
D'	liquid filament diameter (μm)
$\vec{F}_g$	gravity force
$\vec{F}_s$	surface tension force
f	perturbation frequency (Hz)
H_0	nozzle height (μm)
k	wavenumber/ $k = \pi D_0 f / V_m$
L	length of liquid filament (mm)
L_j	jet breakup length (mm)
$\hat{n}$	the unit normal vector of the gas–liquid interface
$\vec{u}$	unit normal vector
Oh	Ohnesorge number
P	pressure (Pa)
Re	Reynolds number
t	time (ms)
$\vec{u}$	speed vector
U	speed
U_j	jet speed (m.s^{-1})

U_0	unperturbed jet velocity (m.s^{-1})
We	Weber number
We_g	Weber number based on the gas density

Greek symbols

ρ	density
ρ_l	liquid density (kg.m^{-3})
ρ_g	gas density (kg.m^{-3})
μ	viscosity ($\text{kg.m}^{-1}.\text{s}^{-1}$)
μ_l	liquid viscosity ($\text{kg.m}^{-1}.\text{s}^{-1}$)
μ_g	gas viscosity ($\text{kg.m}^{-1}.\text{s}^{-1}$)
κ	the curvature of the local gas–liquid interface
σ	surface tension (N.m^{-1})
φ	volume fraction of the liquid phase
ε	modulation amplitude

Abbreviations

LDR	Liquid Droplet Radiator
VOF	Volume-of-Fluid
CSF	Continuous Surface Force
PISO	Pressure Implicit with Splitting of Operation
PRESTO!	Pressure Staggering Option

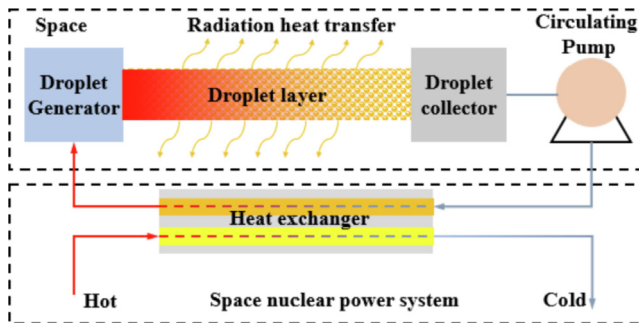


Fig. 1. The layout of the liquid droplet radiator.

transfer is not a concern since the breakup of liquid jets and subsequent initial droplets formation occur during a short traveling time and distance.

Some countries including the United States, Japan, Russia, and China have shown great interest in LDR, and a series of experimental and numerical studies have been conducted to evaluate the LDR working performance. Many experiments were carried out to study the distribution characteristics of droplet stream and droplet size which are very important to evaluate the performance of droplet generators. White (1987) found that the appropriate number of droplet streams is between 10^5 and 10^7 and each droplet stream should contain about 10^5 droplets with a reasonable droplet speed of 3–30 m/s for a megawatt LDR. Hosokawa et al. (1991) investigated the formation characteristics of the liquid droplet stream in a vacuum to develop the droplet generator able to produce uniform droplet streams. Totani et al. (2002, 2005a) studied the geometric and dynamic properties of the droplets based on the operation conditions of the droplet generator and their experiments produced continuous droplet streams with the droplet diameter ranging from 200 to 280 μm under microgravity conditions. Xu et al. (2015) examined the influence of We and Re on the stability of uniform droplet formation by developing a dynamic model for the operation process of the droplet generator. Safronov

(2017) proposed a theoretical model to study the relationship between droplet size and wavenumber which was in good agreement with the experimental results obtained by Bondareva et al. (2019). Xue et al. (2021) carried out experiments to analyze the effects of different operating conditions such as working fluids, ejection pressure, oscillation frequency, nozzle diameter, and the length-diameter ratio of the nozzle on uniform droplet streams.

In addition to the distribution characteristics of droplets, the dynamic features of droplets also play an important role in LDR, which have been always the focus of most attention. In practice, there are usually two types of droplets namely primary droplet and satellite droplet in the breakup process of a laminar liquid jet. The ligament between the primary droplets after the jet breakup eventually becomes spherical and forms the satellite droplets which are usually smaller and at different velocities compared to the primary droplets. For ascertaining the formation conditions of uniform droplet streams, many researchers have studied the droplets formation mechanism and the interactions between them by experiments. Vassallo and Ashgriz (1991) observed four types of satellite droplets formation in the breakup of liquid jets and classified their merging motion modes. Muntz and Dixon (1987, 1989, 1990) found that a special amplitude-modulated disturbance can eliminate the formation of satellite droplets and produce droplet streams with uniform droplet spacing and velocity. Besides, Totani et al. (2002, 2005b) conducted experiments to test the droplet stream collision under a vacuum and microgravity environment and found that the nonuniform droplet stream will cause the droplet splash. Numerical simulations on jet breakup behaviors and droplets formation characteristics have been also performed using various methods. Hilbing and Heister (1996) developed a boundary element method to investigate the nonlinear evolution of the surface of liquid jets and predict the droplet size. They obtained that the reason for the merging of satellites with primary droplets is velocity differences existing between them. Castrejón-Pita et al. (2011) carried out the Lagrangian finite-element simulations to study the influence of the drive amplitude on the jet breakup behaviors and droplet distribution of the liquid jets. Yong et al. (2014) utilized VOF method to examine

the effects of different condition parameters such as oscillation frequency, oscillation amplitude, and Weber number on the jet breakup behaviors. Viswanathan (2018, 2020) investigated the oscillation and coalescence of droplets and found that an air-bubble formation during the droplet merging caused the droplet oscillation along traveling downstream using the VOF method in ANSYS Fluent software. It indicates that the droplet merging and collision caused by satellite droplet formation are destined to destroy the uniformity of the droplet stream.

From the point of view of the jet breakup behaviors, previous studies mostly conducted the single factor analysis and the effects of combined flow and disturbance parameters on the breakup behaviors of a liquid jet are missing, and the effects of Ca on the breakup behaviors of the liquid jet are seldom discussed. Therefore, a comprehensive analysis of the effects of combined flow and disturbance parameters on the breakup behaviors of the liquid jet is necessary. From the point of view of the droplet formation properties, researchers commonly agreed that an LDR requires the formation of uniform droplet streams. Unfortunately, such demand cannot be achieved due to the satellite droplet formation in the breakup process of the liquid jet. Therefore, how to find out the conditions of satellite droplet formation and effectively avoiding satellite droplet formation are very important to ensure the radiation heat transfer performance and continuous operation of an LDR. Even though there is some research about the mechanism of satellite droplet formation, the quantitative evaluation of the satellite formation condition in continuous mode to identify satellite droplets formation region has not been reported. Hence for conveniently and efficiently choosing the appropriate working fluid physical property and operation parameters of droplet generator and improving the radiation heat transfer performance of an LDR, a quantitative criterion for the satellite droplet formation condition in continuous mode should be presented.

In consideration of the practical requirements of an LDR, we paid attention to the satellite droplet formation in the breakup process of a laminar jet in the present study. Extensive numerical simulations were conducted to comprehensively examine the effects of combined flow parameters (We , Oh , Ca) and disturbance parameters (k) on the breakup behaviors of laminar liquid jets and quantitatively analyze the satellite droplet formation regimes in the laminar jet breakup. The objectives of the present study are to carry out investigations on the breakup behaviors of liquid jets and satellite droplet formation characteristics. Besides, quantitative criteria for estimating the satellite formation condition in continuous mode are necessary to be proposed.

2. Physical problem

The breakup behaviors of a laminar liquid jet with continuous droplets formation were studied by imposing a sinusoidal waveform perturbation on the inlet of the nozzle, as shown in Fig. 2. The pressurized fluid with a given mean velocity was ejected from a circular orifice into still air outside the orifice under atmospheric pressure. This paper treated the unperturbed jet velocity (U_0) as the characteristic velocity. The process of droplet formation in a laminar liquid jet breakup is determined by the unperturbed jet velocity (U_0), modulation amplitude (ε), and disturbance frequency (f) of the sinusoidal waveform perturbation. The aspect ratio (H_0/D_0) of the nozzle was fixed at 3.5 in the present study. The vertically oscillating plunger rod caused perturbation which was brought onto the inflow velocity as a sinusoidal waveform. A liquid jet was ejected with the jet velocity (U_j) from the orifice inlet in the direction of gravity, the jet velocity was given by.

$$U_j = U_0(1 + \varepsilon \sin(2\pi ft)) \quad (1)$$

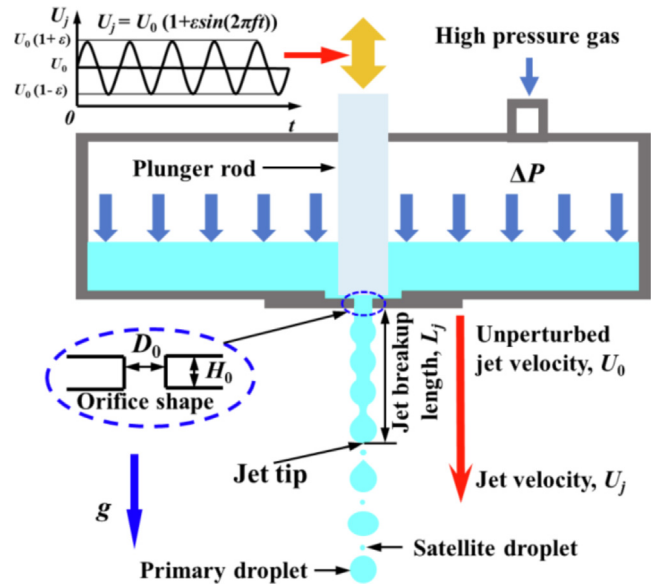


Fig. 2. Schematic diagram of a liquid jet breakup into a continuous droplet stream.

Present simulation conditions are summarized in Table 1. Silicone oil was chosen as the liquid ejected from the nozzle exit with initial jet diameters of 0.1 mm and unperturbed jet velocity of $U_0 = 2\text{--}5$ m/s. Denoting liquid and gas properties with the subscripts l and g , the Other fluid physical properties and disturbance conditions are summarized in Table 1. As defined in Table 1, three independent non-dimensional numbers including Reynolds number (Re), Weber number (We), and disturbance wavenumber (k) used to represent fluid physical properties and operation conditions of droplet generator. In addition, two derived non-dimensional parameters including Ohnesorge number (Oh) and Capillary number (Ca) used to evaluate the stability of liquid jet. These parameters ranges are as follows: Reynolds numbers, $Re = 5\text{--}120$; Weber numbers, $We = 15\text{--}140$; Ohnesorge numbers, $Oh = 0.1\text{--}1.0$; Capillary number, $Ca = 0.5\text{--}7$; disturbance wavenumber, $k = 0.2\text{--}0.8$. Bond numbers for all the simulations are very small, $Bo \sim O(10^{-6})$, which shows that gravity has no impact on the breakup process of a laminar jet, and hence the influence of gravity was negligible in this paper. Fenn and Middleman (1969) suggested that the effect of the atmosphere is possible to be neglected when

Table 1

List of simulation conditions used in the present study.

Items	Values
Liquid	Silicone oil
Gas	Still air
Nozzle radius (D_0)	0.1 mm
Nozzle radius (H_0)	0.35 mm
Unperturbed jet velocity	2–6 m/s
Density of liquid (ρ_l)	920 kg/m ³
Viscosity of liquid (μ_l)	0.0031–0.074 Pa.s
Density of air (ρ_g)	1.225 kg/m ³
Viscosity of air (μ_g)	1.7894×10^{-5} Pa.s
Surface tension of gas–liquid (σ)	0.0123–0.074 N/m
Weber number ($We = \rho_l U_0^2 D_0 / \sigma$)	15–140
Reynolds number ($Re = \rho_l U_0 D_0 / \mu_l$)	5–120
Ohnesorge number ($Oh = \sqrt{We}/Re = \mu_l / (\rho_l D_0 \sigma)^{1/2}$)	0.1–1.0
Capillary number ($Ca = We/Re = \mu_l U_0 / \sigma$)	0.5–7.0
Modulation amplitude (ε)	0.2
Disturbance frequency (f)	2500–10,000 Hz
Disturbance wavenumber ($k = \pi D_0 f / U_0$)	0.2–0.8

$We_g = \rho_g U_0^2 D_0 / \sigma < 5.3$. In our study, all the simulations are within the range, and hence the influence of the atmosphere was not considered likewise.

3. Numerical method

3.1. Governing equations

The breakup process of the liquid jet and droplet formation usually was regarded as a 2D axisymmetric and transient problem (Yong et al., 2014; Li et al., 2020; Viswanathan, 2018, 2020). The axisymmetric time-dependent governing equations were employed to solve continuity and momentum involved in this study and related equations are expressed by:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0, \quad (2)$$

$$\frac{\partial}{\partial t} (\rho \vec{u}) + \nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla p + \nabla \cdot [\mu (\nabla \vec{u} + \nabla \vec{u}^T)] + \vec{F}_g + \vec{F}_s, \quad (3)$$

The VOF model (Hirt and Nichols, 1981) was widely employed to track the gas–liquid interface which was achieved by solving the volume fraction equation and it is given as follows:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{u}) = 0, \quad (4)$$

The volume fraction gives the portion of the cell which is filled with either phase, where $\phi = 0$ the cell is filled with air (the continuous fluid); $0 < \phi < 1$ the interface exists in the cell; $\phi = 1$ the cell is filled with silicone oil (the dispersed fluid). Therefore, the density and viscosity of fluids in the VOF model can be calculated by:

$$\rho = \phi \rho_l + (1 - \phi) \rho_g, \quad (5)$$

$$\mu = \phi \mu_l + (1 - \phi) \mu_g. \quad (6)$$

In this work, the CSF model (Brackbill et al., 1992) was utilized to solve the surface tension force $\vec{F}_s$ in Eq. (3) which is expressed by:

$$\vec{F}_s = \sigma \frac{\rho \kappa \nabla \phi}{\frac{1}{2}(\rho_l + \rho_g)}, \quad (7)$$

Two important physical parameters including the curvature of the local gas–liquid interface (κ) and the unit normal vector ($\hat{n}$) to the interface can be calculated by:

$$\kappa = -(\nabla \cdot \hat{n}), \quad (8)$$

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|} \quad (9)$$

3.2. Boundary conditions and model settings

Considering the axial symmetry of the liquid jet, a two-phase numerical 2D axisymmetric model was assembled by cylindrical

coordinates (x, y), where r is the radial coordinate and x is the axial coordinate in the direction of gravity g respectively, as displayed in Fig. 3. A velocity disturbance with a mean velocity (U_0) imposed on the nozzle orifice was used as the inlet boundary condition. The symmetry condition was provided on the axis and a no-slip boundary condition was implemented at the bottom (wall) and nozzle. A free slip condition was implemented on the side and a pressure-outlet boundary condition was used at the outlet. It was supposed that there should be no mass transfer across the liquid/gas interface and that viscous and surface tension stresses should be in balance. The domain outside nozzle was simplified to a rectangular plane with a width of $5D_0$ and length of $100D_0$ which is suitable for all the cases in the present study.

In the present study numerical simulations were carried out utilizing commercial software package ANSYS Fluent 15.0 based on the finite-volume method. The governing continuum conservation equations for two-phase were solved using a laminar incompressible model integrated with the VOF method of the explicit scheme in a double-precision and transient solver. The PISO arithmetic was chosen to solve the pressure–velocity coupling scheme since it splits the solution into one predictor and two corrector steps which can accelerate the convergence rate in a single iteration step. A second-order upwind scheme was used to discretize the momentum equation in combination with the Least-Squares cell-based method for the discretization of the gradients. The “PRESTO!” (Pressure Staggering Option) scheme was used to interpolate the pressure term. The liquid/gas interface was determined by the volume of fluid (VOF) method using the Geo-Reconstruct Algorithm (Youngs, 1982) based on a second-order piecewise-linear interface construction (PLIC) method. The Continuum Surface Force (CSF) was used to consider the effect of surface tension on the VOF calculation results in a source term in the momentum equation. The first-order implicit scheme conjunction with the fixed time-stepping calculation method was used for time discretization with a maximum Courant number of 0.2 to ensure convergence at each time step. For all residual values, the convergence tolerance achieved within each time-step when monitored was found to be less than 10^{-6} .

3.3. Grid independence and model validation

As shown in Fig. 4, the quadrilateral structured grid was generated using ICFM CFD in our numerical simulations, and the domain is uniformly refined inside the nozzle and along with the travel downstream of liquid.

Before the simulation, grid independence was conducted with $We = 33$, $Oh = 0.21$. The simulation was performed under three meshes with the minimum size of 0.045, 0.05, 0.06 and 0.1 scaled by nozzle diameter corresponding numbers of meshing cells are 53268, 63639, 88149, and 97217, respectively. The jet traveling curves involving the jet growth and jet oscillation was chosen as the indicator and the test results meet the requirements of grid independence. As can be seen from Fig. 5, when the minimum mesh sizes are 0.045, 0.05, and 0.06, they show more similar distributions. In contrast, there is a great difference for the minimum mesh sizes of 0.1. Consequently, the minimum mesh size of 0.05

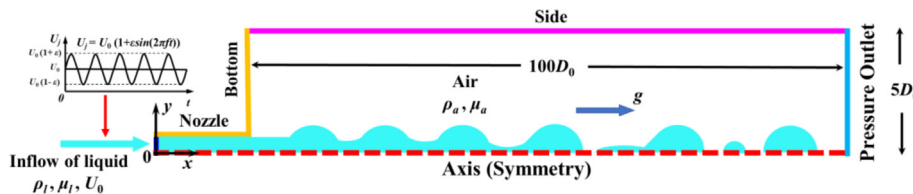


Fig. 3. Schematic of an axisymmetric domain showing boundary conditions, computational domain size used for modeling laminar liquid jet breakup.

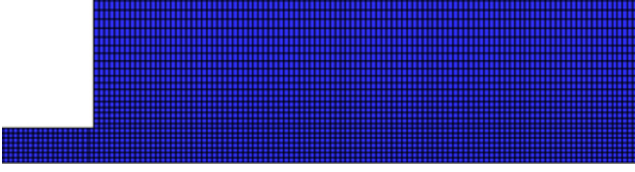


Fig. 4. Computational mesh.

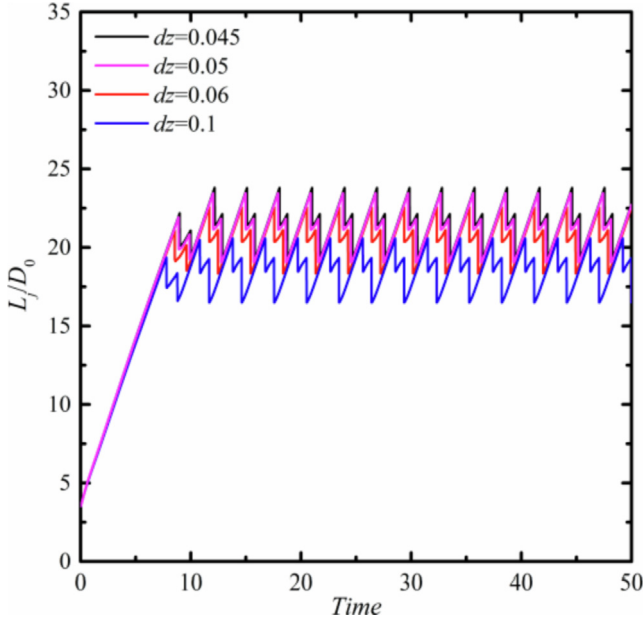


Fig. 5. Grid independence test.

was chosen as the final mesh to reduce the computational resource requirements.

In our present study, CFD results were compared with the analytical (Yuen, 1968) and experimental (Lafrance, 1974; Sallam et al., 2002; Castrejón-Pita et al., 2011) results to validate the simulation model. Fig. 6(a) illustrates the liquid jet shapes for ten

wavelengths behind the breakup point for comparing the results of our simulation and previous experiments (Castrejón-Pita et al., 2011). Reasonably good agreement can be seen for both the droplet shape and jet breakup pattern between the numerical simulations and experiments. As shown in Fig. 6(b), the breakup length was obtained by averaging the instantaneous breakup length over one stable oscillation period of the jet. We compared our simulation results with the correlation summarized by Sallam et al. (2002). However, the calculated jet breakup lengths are shorter than the experimental correlation which can be attributed to the different nozzle shapes. The length-to-diameter ratio of the nozzle chosen in our simulation is much smaller than that used in the experiment. It is known that longer nozzle lengths give rise to longer break-up lengths (McIlroy, 2014). The droplet diameter was obtained by averaging the axial length of near-spherical droplets at the end of the computational domain. Fig. 6(c) shows that the diameter of primary and satellite droplets normalized by the diameter of unperturbed jets as a function of wavenumber agrees well with the experimental results of Lafrance (1974) and the analytical results of Yuen (1968). All the comparisons demonstrate the good agreements between present and previous results which confirm the reliability and accuracy of our numerical simulations for all the cases presented.

4. Numerical simulation results and discussion

4.1. Jet breakup behaviors

In this section, numerical simulations were carried out to examine the effects of combined flow parameters (We , Oh , Ca) and disturbance parameter (k) on the jet breakup behaviors such as the breakup length, breakup time, and breakup pattern of a laminar liquid jet. The parameters ranges are as follows: $We = 20-120$, $Oh = 0.12-0.48$, $Ca = 1-4$, and $k = 0.2-0.8$. The breakup length of the liquid jet, L_j refers to the length of a jet from the orifice exit to the breakup point. The jet breakup time, τ is defined as the evolution time of the jet ejection from the orifice exit to the breakup position, which is normalized with the characteristic timescale of the breakup, $(\rho_l D_0^3 / 8\sigma)^{1/2}$. Moreover, there are two types of breakup patterns namely upstream pinching and downstream pinching during the breakup process of laminar liquid jets, which play an important role in the subsequent droplet formation.

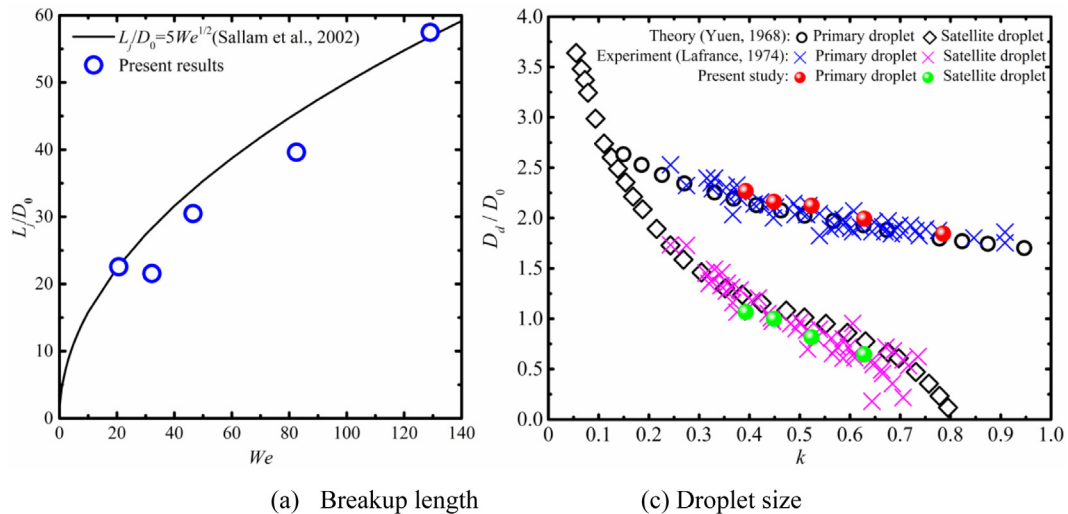


Fig. 6. Comparison of the present study with analytical (Yuen, 1968) and experimental results (Lafrance, 1974; Sallam et al., 2002; Castrejón-Pita et al., 2011) in terms of (a) liquid jet shapes at initial breakup (top and bottom images correspond to the simulated and experimental jets (Castrejón-Pita et al., 2011)); (b) breakup length of laminar jets and (c) diameter of primary as well as satellite droplet.

4.1.1. Effects of Weber number (We) and Ohnesorge number (Oh) on the jet breakup behaviors

Fig. 7 displays the liquid jet profiles at the initial breakup. Each set of jet profiles shows that the liquid jets are with identical We and different Oh numbers. The length of the liquid ligament between adjacent bulges increases with We increased, which may result in the formation of satellite droplets. The liquid ligament gradually becomes thinner due to the capillary force during the liquid jet ejection. Two neck points are occurring between the adjacent bulges and then the formation of primary droplet and satellite droplet is caused by the nonlinear effects. When We is fixed, an increase in Oh can augment the breakup length. Due to the proportional relation between We and velocity, variation in jet velocity will change the value of We , thus causing the instability of the liquid jet. It shows that an increase in We can slenderize and lengthen the liquid ligament between adjacent primary droplets. At high We , the primary droplets present squashed shape owing to the high-speed fluid (when $-\pi/2 + 2n\pi < 2\pi ft < \pi/2 + 2n\pi$) knocking into the lower speed fluid which has been already ejected from the exit of the nozzle. However, as Oh increases, it is observed that this phenomenon is mitigated since the predominant viscous forces can reduce fluid velocity difference in the jets. Therefore, the overall effects of increasing We are to squash the primary droplets and stretch the liquid filament between the adjacent primary droplets, eventually to enhance the instability of the jet. Given the breakup pattern, when $We = 40$, increasing Oh can shift downstream pinching to upstream pinching, which could avoid the satellite droplet formation if the liquid filament is short enough. As We increases, upstream squashed primary droplets cause a large capillary pressure around the upstream neck point, which leads to

the upstream pinching of the liquid filament. However, the squashed primary droplets are absent with Oh increased. Therefore, it is interesting to note that this transition appears reverse when $We = 120$. What all the above analyzed, it can be obtained that high Oh can ensure the liquid jet stable at low We .

Fig. 8 shows the variation of the liquid jet tip position with travel time during the ejection to the initial breakup of liquid jets. The slope of curves can be regarded as the ejection velocity of liquid jets. It can be observed that the ejection of liquid jets during the first perturbation period ($0 \sim 0.2$ ms) is not determined by the viscosity and surface tension of fluid but the inertia. As plotted in Fig. 8(a)–(d), the liquid jet was ejected almost at a constant speed with time before the breakup. Since the higher fluid viscosity further restrain the ejection process, while the identical value of We , the ejection speed of liquid jets was slightly decreased as the values of Oh increased. It proves that the liquid jet ejection process is controlled by both surface tension and viscous force. The breakup time, where the first primary droplet or primary droplet with ligament shed from the liquid jet, also differed depending on the jetting conditions. In Fig. 8, the liquid jet with high Oh showed a slower breakup. Furthermore, as Oh increases by the same amount, it is very interesting to note that at low We (when $We = 20$), the breakup of jets is delayed extremely as compared to other cases. It is known that the breakup length of the liquid jet was governed by both the ejection velocity and breakup time. As shown in Fig. 6, although decreasing Oh can speed up the fluid ejection, the increasing trend of variation of ejection speed is slighter than the decreasing trend of variation of breakup time, which leads to a shorter length of the liquid jet at fixed We numbers.

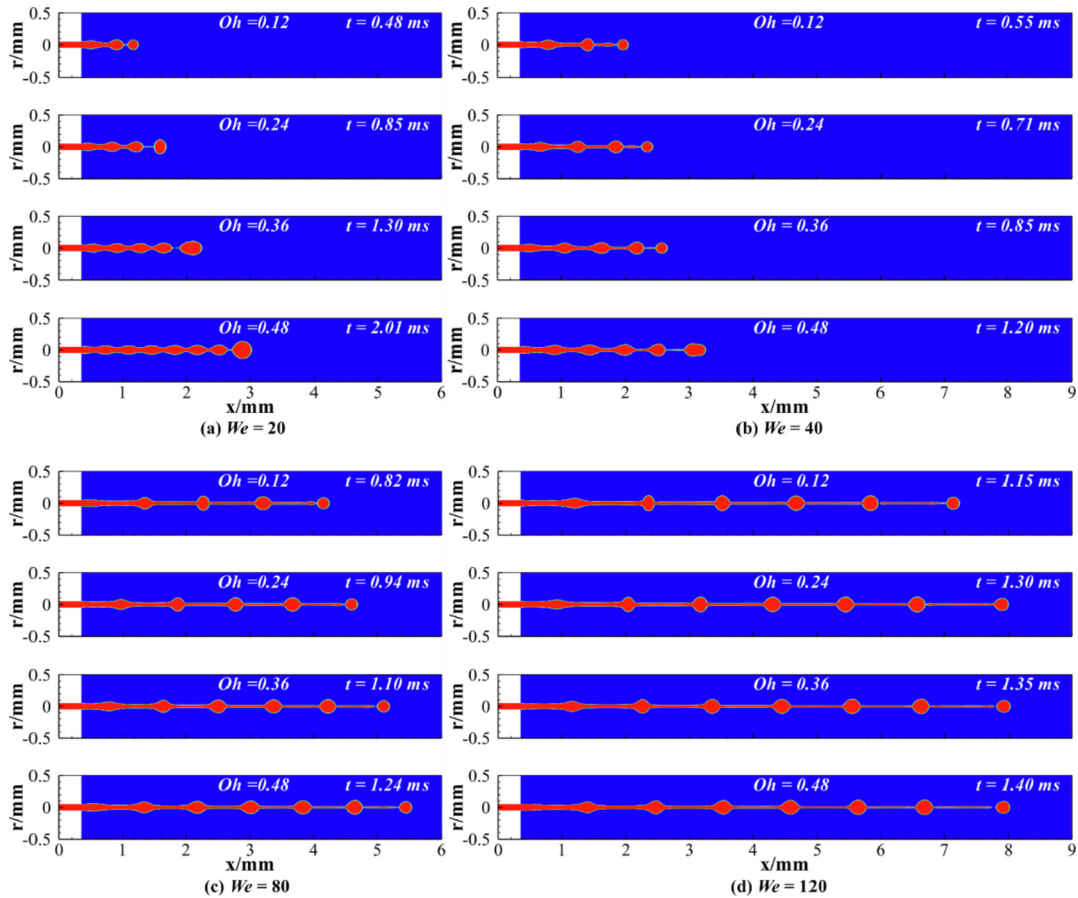


Fig. 7. Effect of Oh on the liquid jet profiles at an initial breakup with $We = 20, 40, 80, 120$.

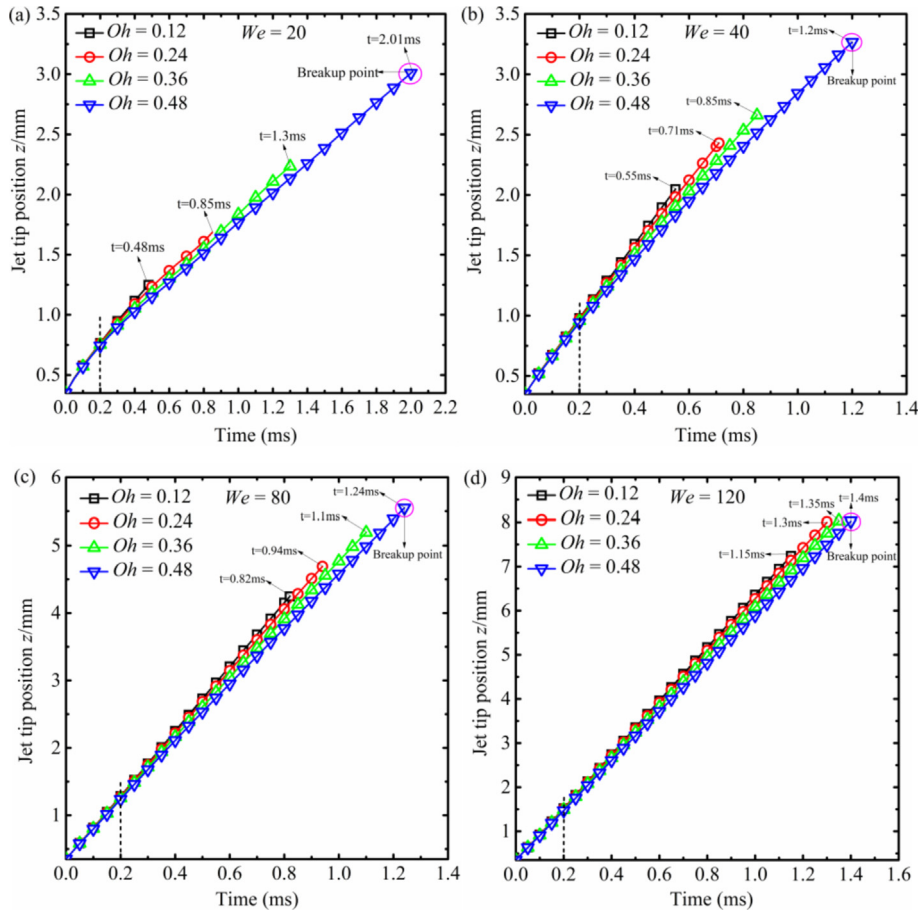


Fig. 8. The variation of the position of jet tip with time from the initial growth to the initial breakup of liquid jets.

Fig. 9 shows the variation of liquid jet tip position with time during an initial breakup to stable oscillation. In these cases, the disturbance frequency of unsteady inflow of liquid was fixed at 5000 Hz, thus one period of perturbation was 0.2 ms. At the initial breakup of liquid jets, the liquid filament between primary droplets will become very thin and its downstream is subjected to a greater surface tension, which leads to the liquid jet tip recoiling. Therefore, it is seen from Fig. 9 that jet tip positions decline after the breakup. Soon afterward, the jet tip position quickly reached a stable oscillation stage. As shown in Fig. 9, the oscillation period of the jet tip is equal to the perturbation period of unsteady inflow of liquid with the value of $\Delta t = 0.2$ ms. Each peak of the oscillation curve denotes the breakup of liquid jet with droplet formation, which means that the generation frequency of droplet is similarly equal to the perturbation frequency of unsteady inflow of liquid. However, as displayed in Fig. 9, the slope of the green dotted line shows a decreasing trend of variation with We decreased, which means that the jet breakup becomes more delayed with Oh increased at low We . Furthermore, the oscillation regimes of liquid jets are also different depending on the jetting conditions. The mean value of liquid jet tip position at oscillation was calculated and plotted via horizontal dotted line in Fig. 9. It shows that the mean value of jet tip position increases with Oh . When $We = 20$, increasing the value of Oh , the mean value of jet tip position increased shapely at $Oh = 0.48$. Furthermore, it can be also found that the oscillation amplitude becomes larger with We increased. The oscillation amplitude of the jet tip depends on the breakup pattern of a liquid jet. Corresponding to what is shown in Fig. 7, it can be seen from Fig. 9 that upstream pinching causes a larger oscillation amplitude of jet, inversely downstream pinching leads

to a smaller oscillation amplitude. This can be also attributed to the relative magnitudes of surface tension at liquid jet tip caused by different breakup patterns.

4.1.2. Effects of capillary number (Ca) and disturbance wavenumber (k) on the jet breakup behaviors

Fig. 10 shows the liquid jet profiles at initial breakup for four different disturbance wavenumbers $k = 0.2, 0.4, 0.6$, and 0.8 . All the liquid jets were ejected from the nozzle exit at $We = 20, 40$, respectively. As described in both Fig. 10(a) and (b), a decrease in Ca enables the breakup of liquid jet advanced which can be explained by the dominance relationships between the capillary and viscous forces. The necking occurs before the jet breakup, capillary forces promote the necking of liquid jet, whereas the effects of viscous forces just reverse. Therefore, the breakup of the liquid jet becomes faster as the Ca decreases. Furthermore, increasing wavenumber will decrease the sizes of primary droplets and liquid ligaments. As shown in Fig. 10, the breakup time and breakup length of liquid jets both decrease at first and then increase at low We with disturbance wavenumber increased, whereas a monotonically decrease at high We .

The term of $(\rho_l D_0^3 / 8\sigma)^{0.5}$ has been considered as the characteristic timescale of jet breakup. As plotted in Fig. 11, the breakup time was nondimensionalized as $\tau = t / (\rho_l D_0^3 / 8\sigma)^{0.5}$, as a function of the disturbance wavenumber. It can be found from Fig. 11(a) and (b) that the breakup time of liquid jet decreases at first and then increases with increasing wavenumber at low We (when $We = 20, 40$). Furthermore, as Ca increases, Fig. 11(a) shows a significant increase in breakup time. The reason can be explained that increasing Ca causes an increase in the viscous forces which can

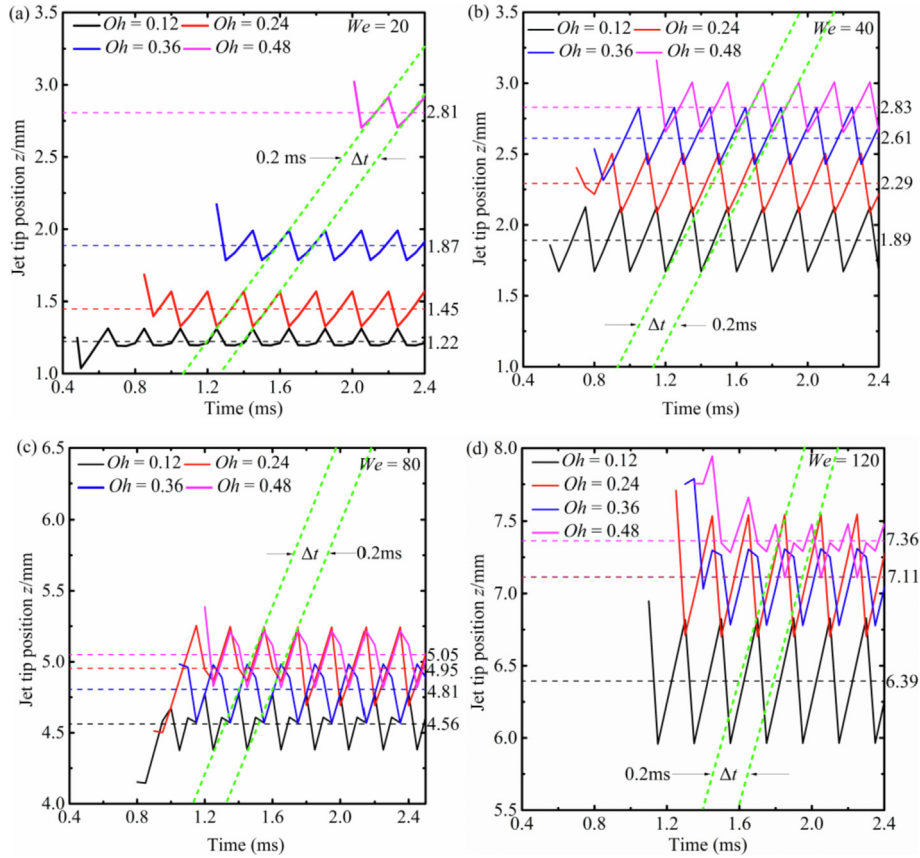


Fig. 9. The variation of the position of jet tip with time from the initial breakup to stable oscillation.

restrain the ejection process of liquid jets. The enhanced viscous forces make sure the liquid jet stability so that the breakup of liquid jets is further delayed. However, as shown in Fig. 11(c) and (d), the breakup time takes on a monotonous decreasing trend of variation with wavenumber at high We (when $We = 80, 120$) which shows a deviation from Rayleigh's theory. It can be attributed to the stronger nonlinear interactions caused by increasing Weber numbers. It is worth noting that the effects of Ca on the breakup time were not dominant for all wavenumbers with We increased.

The mean breakup length, L_j is one of the most significant properties to characterize the breakup behaviors of liquid jets, which is closely linked to the straight jet required for the practical applications in LDR. It is a fact that a shorter jet can ensure favorable straightness and a faster pinch-off. The mean breakup length is calculated by averaging the instantaneous value of the breakup length over one stable oscillation period of the liquid jet. As disturbance wavenumber increases, it is as expected to observe from Figs. 11 and 12 that there is a close correspondence between the breakup length and breakup time for variation trend. Whereas the breakup length of liquid jet decreases at first and then increases with increasing wavenumber only at low We and high Ca . As We increases, a great change appears in the trend of variation of breakup length which shows a monotonous decrease in breakup length for all wavenumbers. The nonlinear effects on the liquid jet have been also observed in previous experiments (Meier et al., 1992). Furthermore, it is surprised to find that increasing Ca only slightly affects breakup length at high We (when $We = 80, 120$) for all wavenumbers.

In quantitatively describing jet breakup behaviors, one usually refers to the behavior of the continuous part of the jet, or breakup length. Through the previous study, it is known that the breakup

length for a laminar liquid jet can be derived from the classical linear instability theory proposed by Weber (1931) which is expressed by.

$$\frac{L_j}{D_0} = C(We^{1/2} + 3Ca) \quad (10)$$

However, due to the failure for directly predicting the jet breakup length via Eq. (10), the model of breakup length of a laminar liquid jet was corrected by Grant and Middleman (1966) as:

$$\frac{L_j}{D_0} = C(We^{1/2} + 3Ca)^{0.85} \quad (11)$$

where the non-deterministic term C was provided as $C = A - B \ln(Oh)$, and the A, B are determined by operation conditions of the liquid jet ejection.

For conveniently calculating the breakup length of a laminar liquid jet from the operation conditions and facilitating to design of a good droplet generator for LDR. It is necessary to develop a new empirical equation for the breakup length of a laminar jet that suffered a sinusoidal waveform perturbation. According to the results discussed above, we can know that the breakup behaviors of a laminar liquid jet are controlled by We, Oh, Ca , and k . It is usually assumed that the imposed perturbation takes on an exponential variation in space. Moreover, it has been obtained that the capillary number only has a significant impact on the breakup behaviors at low We . Consequently, the following empirical equations were established by using the above simulation results:

When the liquid jet ejected at low We (when $We = 20, 40$):

$$\frac{L_j}{D_0} = [9.34 - 0.86 \ln(Oh)](We^{0.5} + 3Ca)^{0.82} \cdot \exp(50h.k^3 - 3k) \quad (12)$$

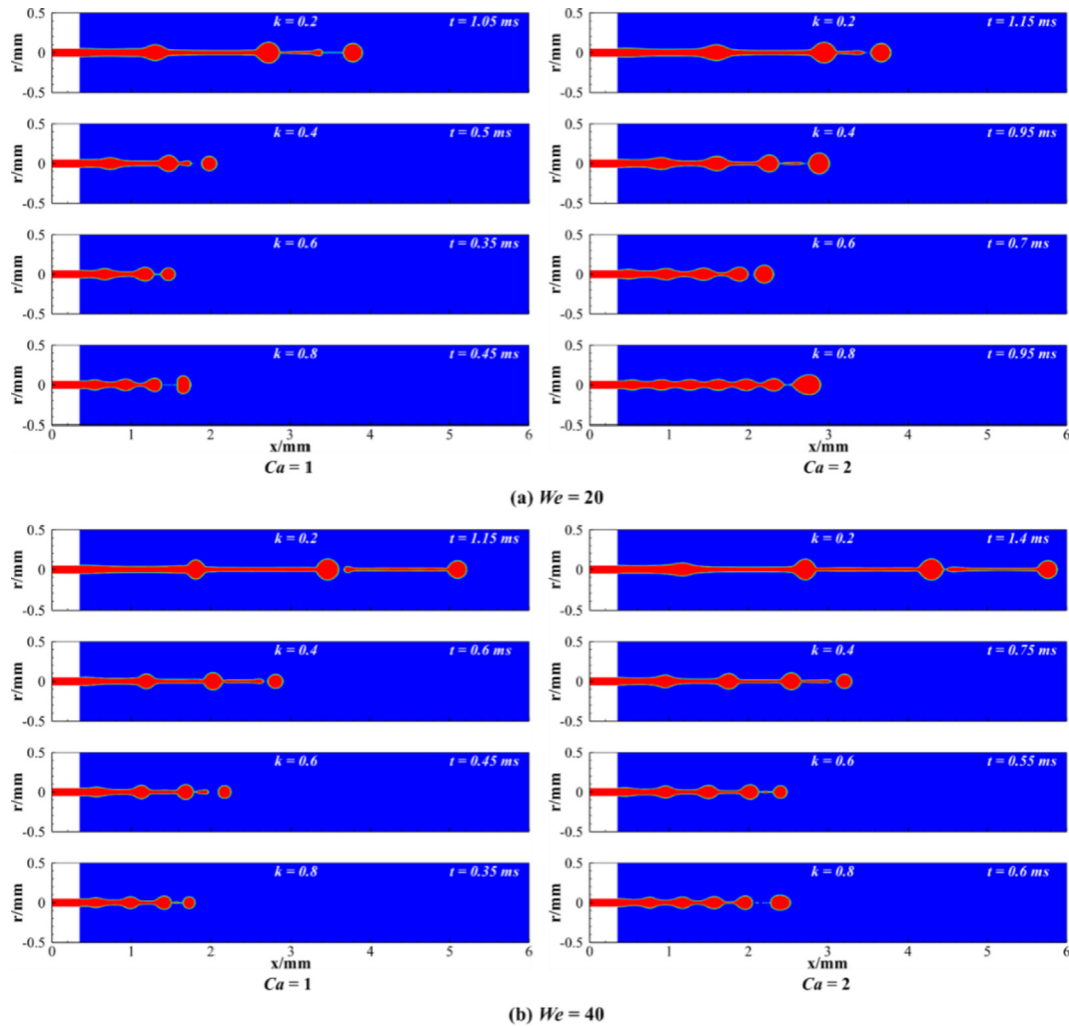


Fig. 10. Effect of k on the liquid jet profiles at an initial breakup with $Ca = 1, 2$ under $We = 20, 40$.

When the liquid jet ejected at high We (when $We = 80, 120$):

$$\frac{L_j}{D_0} = [9.34 - 0.86 \ln(Oh)](We^{0.5} + 3Ca)^{0.82} \cdot \exp(2k^3 - 3k) \quad (13)$$

Fig. 12 also shows the comparison of laminar liquid jet breakup length between empirical equations calculation and numerical simulation. The results indicate that the calculated values by empirical equations agreed well with the simulation results. The maximum deviation was about 26% with k ranging from 0.2 to 0.8.

4.2. Droplet formation properties

In this section, numerical simulations were carried out to study the effects of satellite droplet formation on uniform droplet streams. Moreover, we made efforts to quantitatively describe the satellite droplet formation regimes. The parameters ranges are as follows: $We = 15-140$, $Oh = 0.1-1.0$, and $k = 0.2-0.8$, which are within the appropriate ranges for LDR.

4.2.1. Effects of satellite droplet formation on uniform droplet stream

Fig. 13(a) shows the jet profiles at three moments near a droplet pinch off the liquid filament for $We = 40$, $Ca = 1$, and $k = 0.6$. Fig. 13(b) shows a plot of axial velocity at the centerline of satellite and primary droplets. For finite-length liquid jets, droplets are shed from the liquid jet sequentially so that the droplet velocities are

not necessarily equal. The simulations presented here show that the satellite droplets are shed from the jet with velocities less than that of the primary droplets. Each profile in Fig. 13(a) has been shifted axially so that the left-hand side of the jet remains in the same relative location. Just after a primary droplet is shed from the jet, the surface of the satellite droplet is approximately conical at the jet tip. As noted in Fig. 13(a), the rebound distance of the jet tip is increased before the satellite droplet is completely shed from the jet. Surface tension acts to pull this liquid into a spherical shape and tends to slow the velocity of the satellite droplet as seen in Fig. 13(b). It is explained that satellite droplets of constant axial velocity are slightly less than that of the primary droplets due to the breakup pattern of downstream pinching.

It is known that the mode of interaction between droplets depends on the jet breakup patterns. The simulations presented here show a downstream pinching for liquid jet breakup which causes rear coalesced satellite droplets formation in the droplet stream. Based on the discussion above, it is can be obtained that an upstream pinching for liquid jet breakup could cause forward coalesced satellite droplets formation in the droplet stream.

As shown in Fig. 14, it is respectively displayed that satellite droplet is shedding from the primary droplet and axial velocity at the centerline of satellite and primary droplets. The breakup pattern of liquid jets is downstream pinching as seen in Fig. 14(a). The satellite droplet forms followed by the primary droplet. The axial

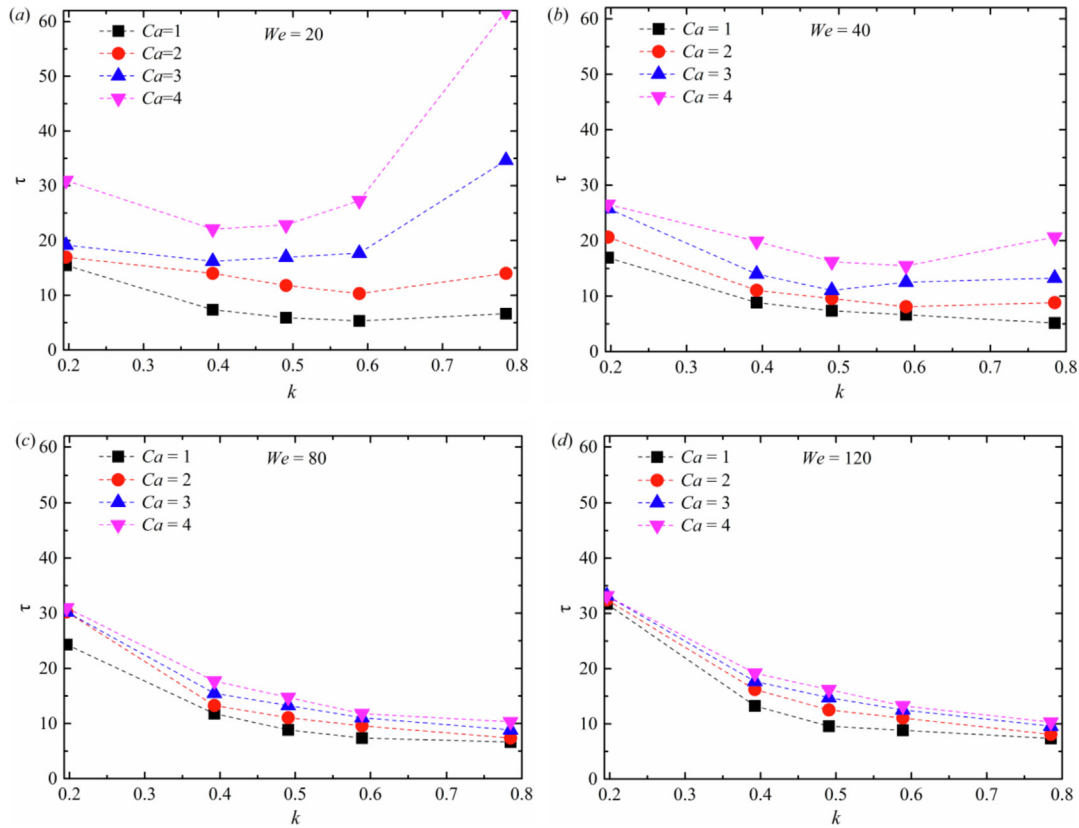


Fig. 11. Effect of Ca on breakup time for the disturbance wavenumber range of 0.2–0.8 under $We = 20, 40, 80, 120$.

velocity of the center of mass of the droplet was considered as the droplet velocity. It can be observed in Fig. 14(b) that the satellite droplet velocity is less than that of the primary droplet, which means the next primary droplet will chase after the previous satellite droplet until coalesced with each other. For this case, the velocity difference between them is equal to 0.03 m/s. However, the spacing between adjacent satellite and primary droplets is approximately equal to 0.3 mm. It is obtained that the adjacent satellite and primary droplets will have rear coalesced for a short distance downstream 10 ms later. For the application of LDR, the length of a typical rectangular droplet layer is 50 m, and hence the flight time of the droplet is 12.5 s with a droplet velocity of 4 m/s. It is known that the uniform droplet stream required for LDR should contain a series of droplets of the same size and velocity. It indicates that for practical application, the satellite droplets formation in the breakup process of liquid jets will cause droplets to coalesce and destroy uniform droplet streams.

4.2.2. Criteria for satellite droplets formation

Given the uniformity of droplet formation, it overwhelmingly depends on whether there is satellite droplets formation in the liquid jet breakup process. Hence, we proposed quantitative criteria for the satellite droplet formation condition in continuous mode to identify the satellite droplet formation region under different jetting conditions. Fig. 15 illustrates a series of continuous droplet streams formation at the steady stage, which presents three types of droplet formation regimes including no satellite droplet formation, single satellite droplet formation, and multiple satellites droplet formation. The information in the figure shows that increasing We will stretch the liquid filament between adjacent primary droplets which may lead to one or more satellite droplets formation. It can be observed from the figure that when We is identical, increasing Oh can thin the liquid filament between primary droplets, and

the satellite droplets gradually become much smaller until no satellites form at larger Oh .

An LDR has high demands on uniformity of drop size for achieving a good radiation heat transfer performance. However, such a desire usually falls through due to the existence of satellites even sub-satellite droplets. Therefore, it is vital to identify the satellite droplet formation regions under the different fluid physical properties and operation conditions of the droplet generator. Fig. 16 provides relationships between the different flow parameters in terms of Oh and We at the range of $15 < We < 140$, $0.1 < Oh < 1.0$ which is suitable for a laminar jet breakup in LDR. The black solid circles represent no satellite droplet generation, the green solid triangles and the red solid squares signify the formation of the single droplet and multiple satellites droplet, respectively. The boundaries of satellite droplet formation regions plotted by the fitted curves are shown in Fig. 16. An important observation made from the phase diagram is that when We is the same, increasing Oh will shift the single or multiple satellites region to a no-satellite droplet region. The result shows that a liquid jet with higher viscosity will lead to a slower breakup and prevent satellite droplet formation. It also suggests that much higher liquid viscosity is required for preventing the satellite droplet formation for a jet with low surface tension.

It can be seen from our simulation results that the initial liquid filament shed from the liquid jet likely generates three types of satellite droplet regimes, as illustrated in Fig. 17. Researchers (Castrejón-Pita et al., 2012; Hoath et al., 2013) have tried to identify the boundaries of satellite droplet formation regimes by considering two non-dimensional numbers, such as the Ohnesorge number and the aspect ratios of liquid filament, L/D' .

Castrejón-Pita et al. (2011) experimentally and computationally studied the droplet formation in continuous mode and found that the process of droplet formation in a laminar liquid jet breakup is

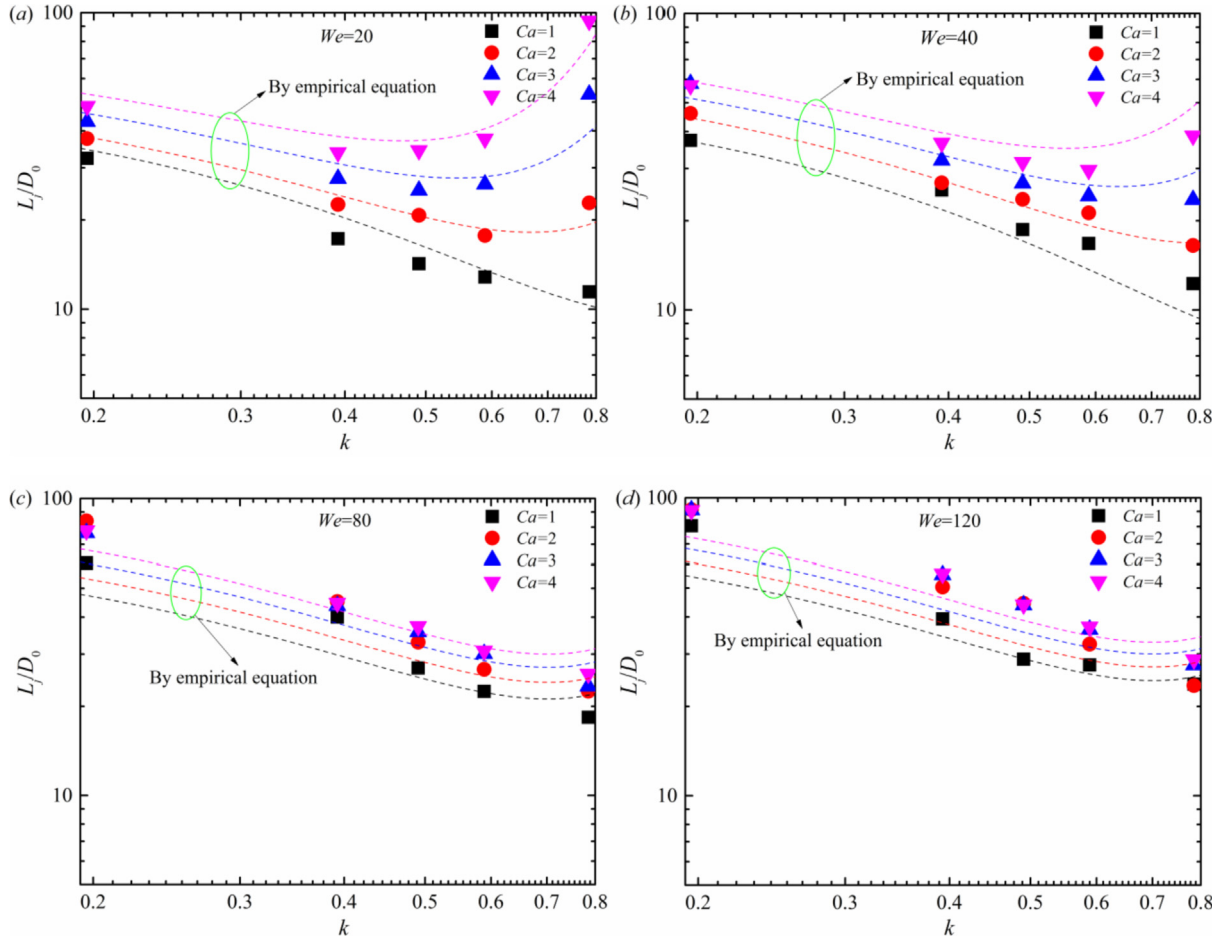


Fig. 12. Effect of Ca on breakup length for the disturbance wavenumber range of $0.2 \sim 0.8$ under $We = 20, 40, 80, 120$.

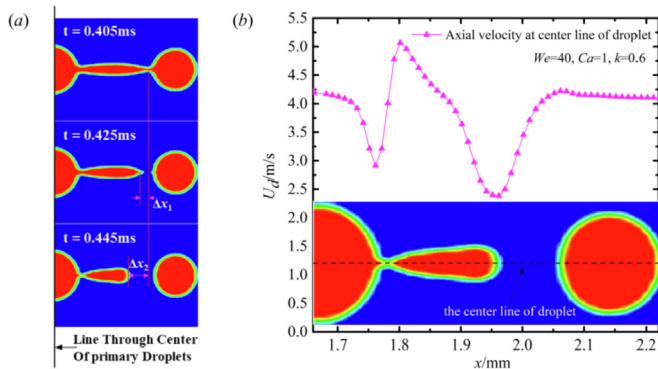


Fig. 13. (a) Satellite droplet evolution just before shedding; (b) axial velocity at the centerline of droplet before the satellite droplet is separated from the calculation (at $t = 0.445$ ms).

controlled by liquid physical properties, nozzle geometry, and disturbance condition. Hence, the current study uses the nozzle diameter and the jet speed to represent the nozzle geometry and disturbance condition, and the size of liquid filament can be respectively expressed by:

$$L = f(\rho, \sigma, \mu, V, D), D' = f(\rho, \sigma, \mu, V, D) \quad (14)$$

The scaling analysis technique (Emori et al., 2000) was employed to transform the above equations and the following is derived:

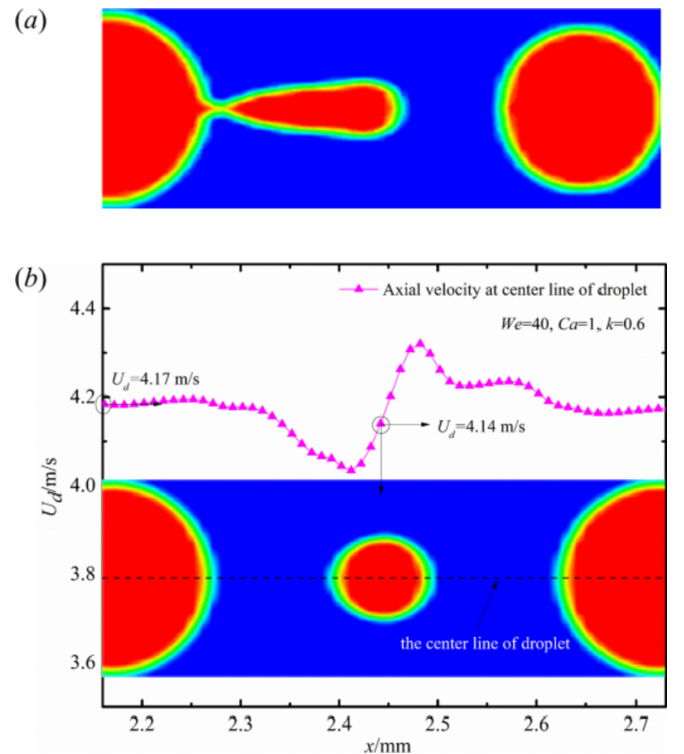


Fig. 14. (a) Satellite droplet is shedding from a primary droplet; (b) axial velocity at the centerline of satellite and primary droplets.

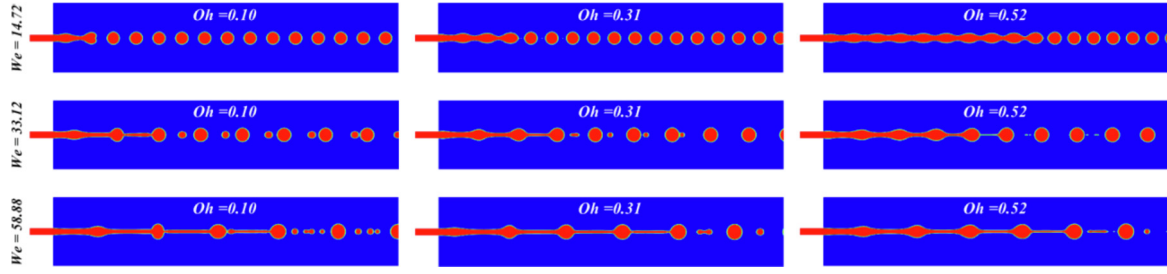


Fig. 15. Three types of droplet formation regimes: a) no satellite droplet formation; b) single satellite droplet formation; c) multiple satellites droplet formation.

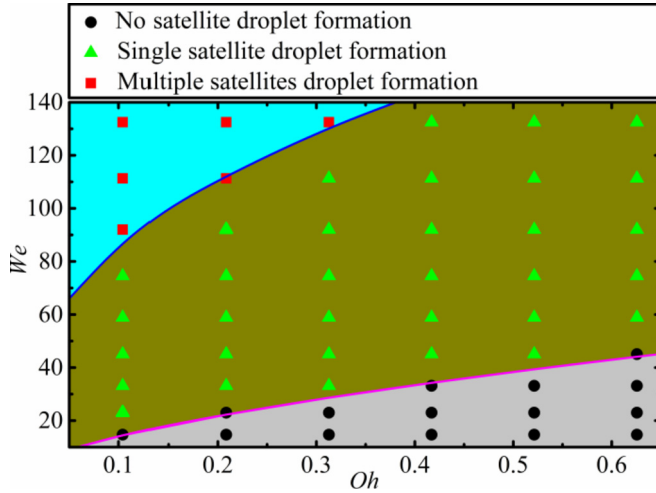


Fig. 16. Phase diagram of satellite droplet formation regimes in the Oh - We domain.

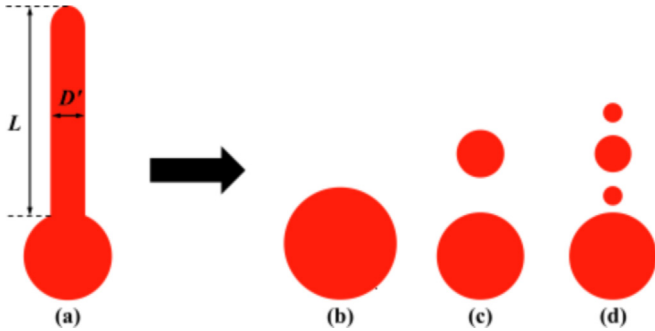


Fig. 17. Schematic diagram of (a) initial liquid filament (b) no satellite droplet (c) single satellite droplet (d) multiple satellites droplet.

$$L/D = f(We, Re), D'/D = f(We, Re) \quad (15)$$

and then.

$$L/D' = f(We, Re) \quad (16)$$

Therefore, the aspect ratio of liquid filament is a function of We and Re . Although Oh , Re , and We were used to describe the process of droplet formation in a laminar liquid jet breakup, one of (Re and We) and (We and Oh) can achieve delineating the boundaries between the two regimes due to their interdependency. Zhan et al. (2020) classified the breakup regime of the liquid jet by formalizing the boundary with Re as a power function of Oh . It is a fact that the droplet formation is directly related to the breakup behaviors of the liquid jet, thus we similarly used power function of We and Oh to formalize the satellite droplet formation regime boundary. The criterion for no satellite droplet formation regime is expressed by.

$$\frac{We}{590Oh^{0.62}} < 1 \quad (17)$$

The criterion for multiple satellites droplet formation regime is given by.

$$\frac{We}{2000h^{0.37}} > 1 \quad (18)$$

and the single satellite droplet formation regime is between them, as follows:

$$\frac{We}{590Oh^{0.62}} > 1, \text{ and } \frac{We}{2000h^{0.37}} < 1 \quad (19)$$

To examine the accuracy of the criteria above, a comparison of the satellite droplet formation regime that is obtained from simulation results and previous experiment results (Vassallo and Ashgriz, 1991) is presented in Table 2, which suggests that our proposed criteria can well identify the satellite droplet formation regime.

As mentioned above, the aspect ratio of liquid filament has a function of Ohnesorge number during the droplet formation. Furthermore, the size of liquid filament has an important influence on the droplet size which indicates that the droplet size is related to Oh . Considering the mass conservation, the mass of the cylindrical liquid jet in one wavelength is equal to the mass of the spherical droplet after the jet breakup, which can be stated as follows:

$$\pi \frac{D_j^2 \lambda}{4} = \pi \frac{D_d^3}{6} \quad (20)$$

According to the relationship between wavelength and wavenumber, $k = \pi D_j / \lambda$, Eq. (16) can be transformed into:

$$\frac{D_d}{D_j} = \left(\frac{1.5\pi}{k} \right)^{1/3} \quad (21)$$

And $Oh = \frac{\mu}{\sqrt{\rho \sigma D_j}}$, it can be derived that.

$$D_d = COh^{-2} k^{-1/3} \quad (22)$$

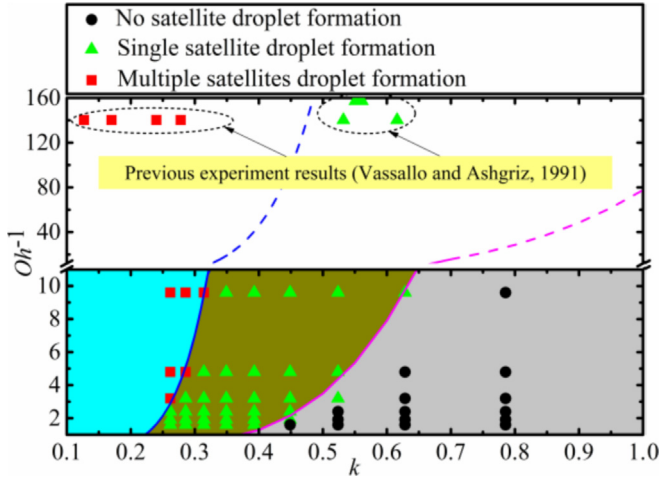
where D_j , D_d is the jet diameter and droplet diameter, respectively; C is a constant.

Thus, Oh and k can be also used to describe the process of droplet formation in a laminar liquid jet breakup, and we similarly used power function to formalize the satellite droplet formation regime boundary in terms of Oh and k for associating the fluid physical properties with the operation condition of droplet generator. Fig. 18 provides a phase diagram of satellite droplet formation regimes in terms of Oh and k at the range of $0.2 < k < 0.8$, $0.1 < Oh < 1.0$ which is suitable for a laminar jet breakup in LDR. Two boundaries are plotted via two different colored curves to identify the satellite droplet formation regimes. It is observed from Fig. 18 that the slope of the red curve denoted the boundary of satellites/no-satellite shows a slowly increasing trend of variation at first and then increases rapidly after $k = 0.7$. The result indicates

Table 2

Comparison of our prediction and experiment results.

Items	Values									
Experimental results	We	41.2	41.2	41.3	41.2	30.6	30.6	30.7	27.3	27.3
	Oh	0.0071							0.0064	
	N	4	3	2	2	1	1	1	1	1
Predicted results	$\frac{We}{2000Oh^{0.37}}$	1.28	1.28	1.29	1.28	0.95	0.95	0.96	0.89	0.89
	$\frac{We}{590Oh^{0.52}}$	2.33	2.33	2.33	2.33	2.01	2.01	2.01	2.04	2.04
	Satellite droplet regime	$\frac{We}{2000Oh^{0.37}} > 1$				$\frac{We}{590Oh^{0.52}} > 1$, and $\frac{We}{2000Oh^{0.37}} < 1$				
		Multiple	Multiple	Multiple	Multiple	Single	Single	Single	Single	Single

Note: where N represents the number of satellites droplet formation in experiments (Vassallo and Ashgriz, 1991).**Fig. 18.** Phase diagram of satellite droplet formation regimes in the k - Oh domain.

that satellite droplet is difficult to exist in the breakup of liquid jet with a short-wavelength disturbance. However, the satellite droplets are eventually formed in the breakup of a liquid jet with a long-wavelength disturbance, which only disappears for very larger Oh values. Therefore, much larger Oh jets are required for restraining satellite droplet formation. However, when the disturbance wavenumber is less than 0.4, satellite droplets are always formed in a jet breakup, and much more satellite droplets may be about to form in a jet breakup when the disturbance wavenumber is less than 0.2. Likewise, a comparison of the satellite droplet formation regime that is obtained from simulation results and previous experiment results was also made to examine the accuracy of our simulation results. Although the experimental operation range is beyond our simulation range, our criteria can also well predict the experiment results.

Finally, the criterion for no satellite droplet formation is expressed by.

$$\frac{Oh^{-1}}{78k^{4.48}} < 1 \quad (23)$$

The criterion for multiple satellites droplet formation regime is given by.

$$\frac{Oh^{-1}}{17800k^{6.51}} > 1 \quad (24)$$

and the single satellite droplet formation regime is between them, as follows:

$$\frac{Oh^{-1}}{78k^{4.48}} > 1, \text{ and } \frac{Oh^{-1}}{17800k^{6.51}} < 1 \quad (25)$$

5. Conclusions

As a frameless heat removal system of space nuclear power system, the liquid droplet radiator (LDR) has received increasing attention. In consideration of the practical requirements of an LDR, satellite droplet formation in the breakup process of a laminar liquid jet was numerically investigated by imposing a sinusoidal waveform perturbation at the operation range of $15 < We < 140$, $0.1 < Oh < 1.0$, $0.5 < Ca < 7$ and $0.2 < k < 0.8$. The effects of combined flow and disturbance parameters on the breakup behaviors of laminar jets were examined and the satellite droplet formation regimes in laminar liquid jets breakup were quantitatively analyzed. The main conclusions are summarized as follows:

- (1) The effects of flow parameters (Oh , We) on the jet breakup behavior were examined. When $We = 40$, increasing Oh can shift the breakup pattern of a liquid jet from downstream pinching to upstream pinching, but this transition appeared reverse when $We = 120$. The liquid jet breakup was greatly delayed with Oh increased at low We and the oscillation amplitude of liquid jet tip became larger with We increased. In summary, increasing We can enhance the instability of liquid jet and high Oh can stabilize the liquid jet at low We .
- (2) The effects of combined flow parameters (We , Ca) and disturbance parameter (k) on the liquid jet breakup behaviors were analyzed. At low We , the breakup length and breakup time of liquid jets both presented a decreasing followed by an increasing trend of variation with k . Whereas they have a monotonously decreasing trend of variation with k at high We . On the other hand, an increase in Ca significantly increased the breakup length and breakup time of liquid jet at low We , but it had a slight influence on the breakup behaviors at high We .
- (3) The effects of satellite droplet formation on uniform droplet streams were discussed. It was verified that the droplets coalesced downstream due to their velocity difference. A downstream pinching for liquid jet breakup would cause rear merger satellite droplets formation in the droplet stream and an upstream pinching for liquid jet breakup could cause forward merge satellite droplets formation in the droplet stream. Therefore, for practical application of LDR, the satellite droplets formation in the breakup process of the liquid jet would destroy uniform droplet streams.
- (4) Two visual phase diagrams were obtained to describe the satellite droplet formation regimes in the breakup process of laminar liquid jets. Quantitative criteria were proposed to estimate the satellite formation condition in continuous mode and criteria were formalized with Oh and We to identify three types of satellite droplet formation regimes. Besides, marginal jet Oh numbers for a range of disturbance

wavenumbers of 0.2–0.8 were provided to identify the satellite droplet formation regime and our criteria and phase diagram well predicted the previous experimental results.

CRedit authorship contribution statement

Maoguo Zhu: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing – original draft. **Quanbin Zhao:** Methodology, Validation, Writing – review & editing, Supervision. **Daotong Chong:** Resources, Methodology, Validation, Writing – review & editing, Supervision, Project administration. **Weixiong Chen:** Writing – review & editing, Methodology. **Junjie Yan:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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